VLSI DESINING INTERNSHIP

TASK-2

INTRODUCTION

Verilog is a Hardware Description Language (HDL) used to design and simulate digital circuits. In this task, basic logic gates, a Half Adder, and a Full Adder were designed using Verilog. The designs were simulated using Icarus Verilog on EDA Playground, and the outputs were verified using EPWave.

OBJECTIVE

To design and simulate basic logic gates,Half adder and Full adder using Verilog.The outputs are verified through simulation waveforms using EPWave.

TOOLS USED

Verilog – Hardware Description Language used for circuit design.

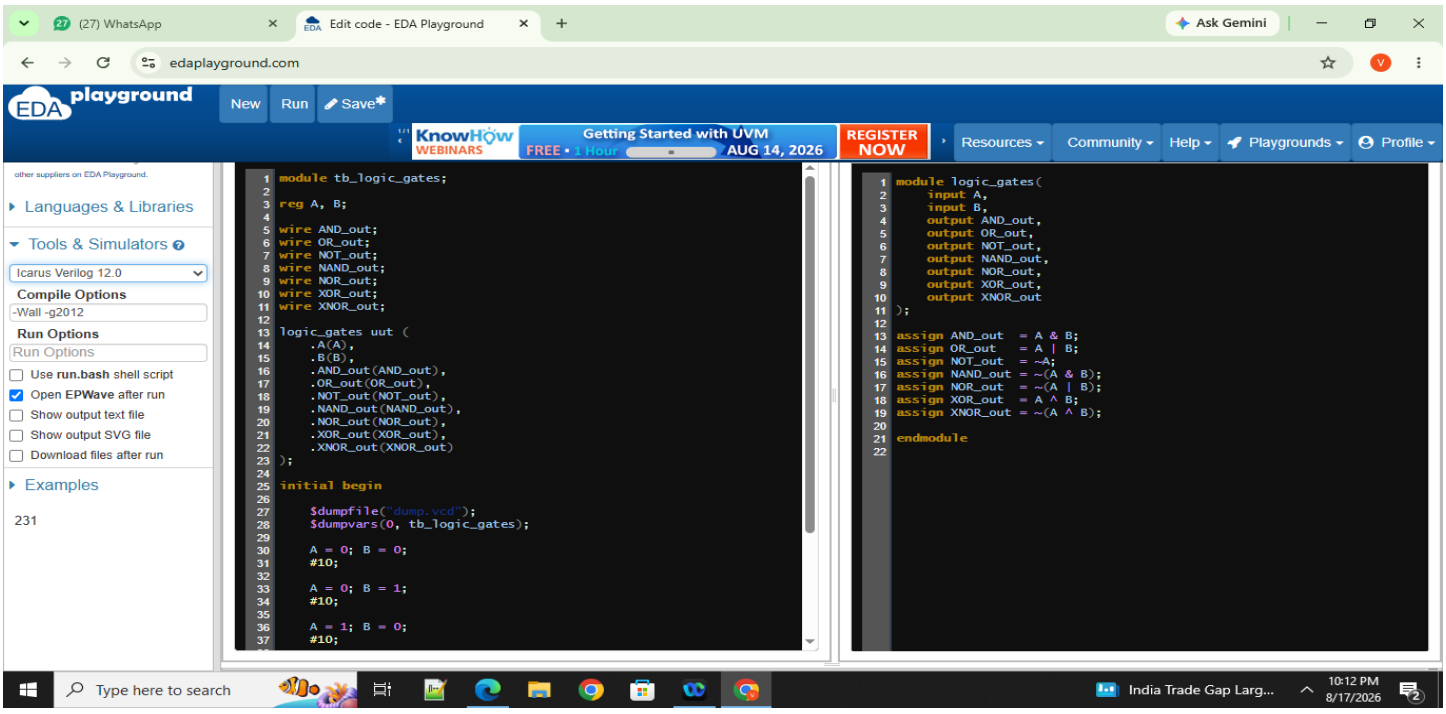
Icarus Verilog – Used to compile and simulate the Verilog code.

EDA Playground – Used as the online simulation environment.

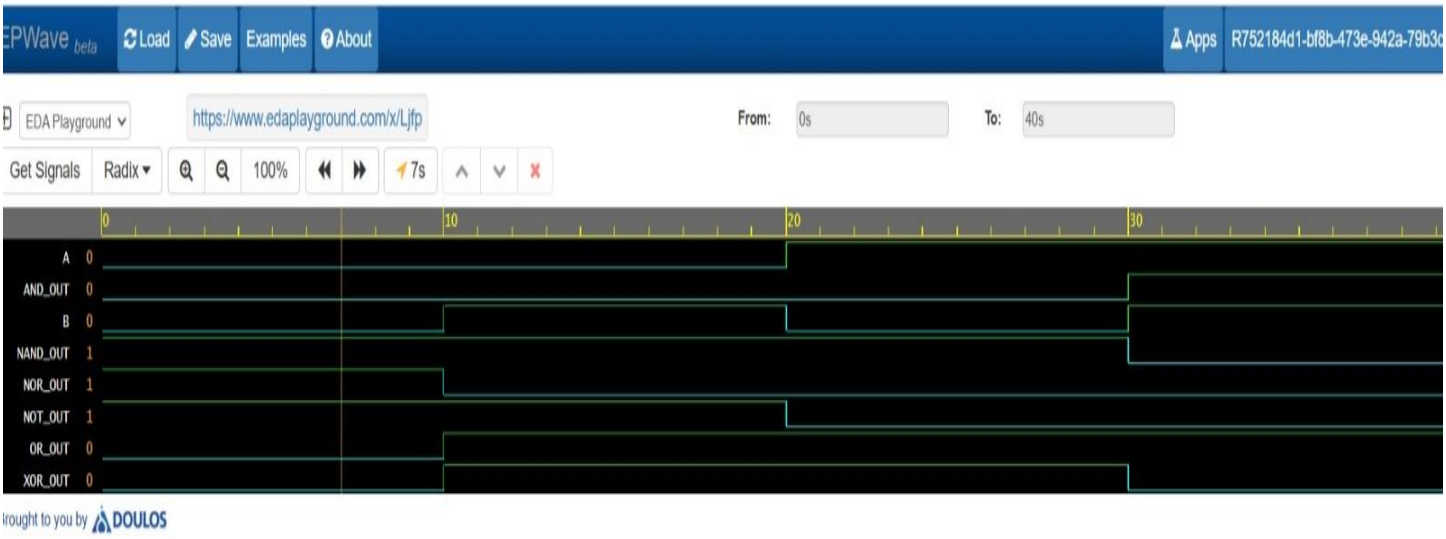
EPWave – Used to view and analyze the simulation waveforms.

LOGIC GATES

The logic gates were implemented using Verilog HDL, and their functionality was verified through simulation using a testbench. The implemented gates include AND, OR, NOT, NAND, NOR, XOR, and XNOR.

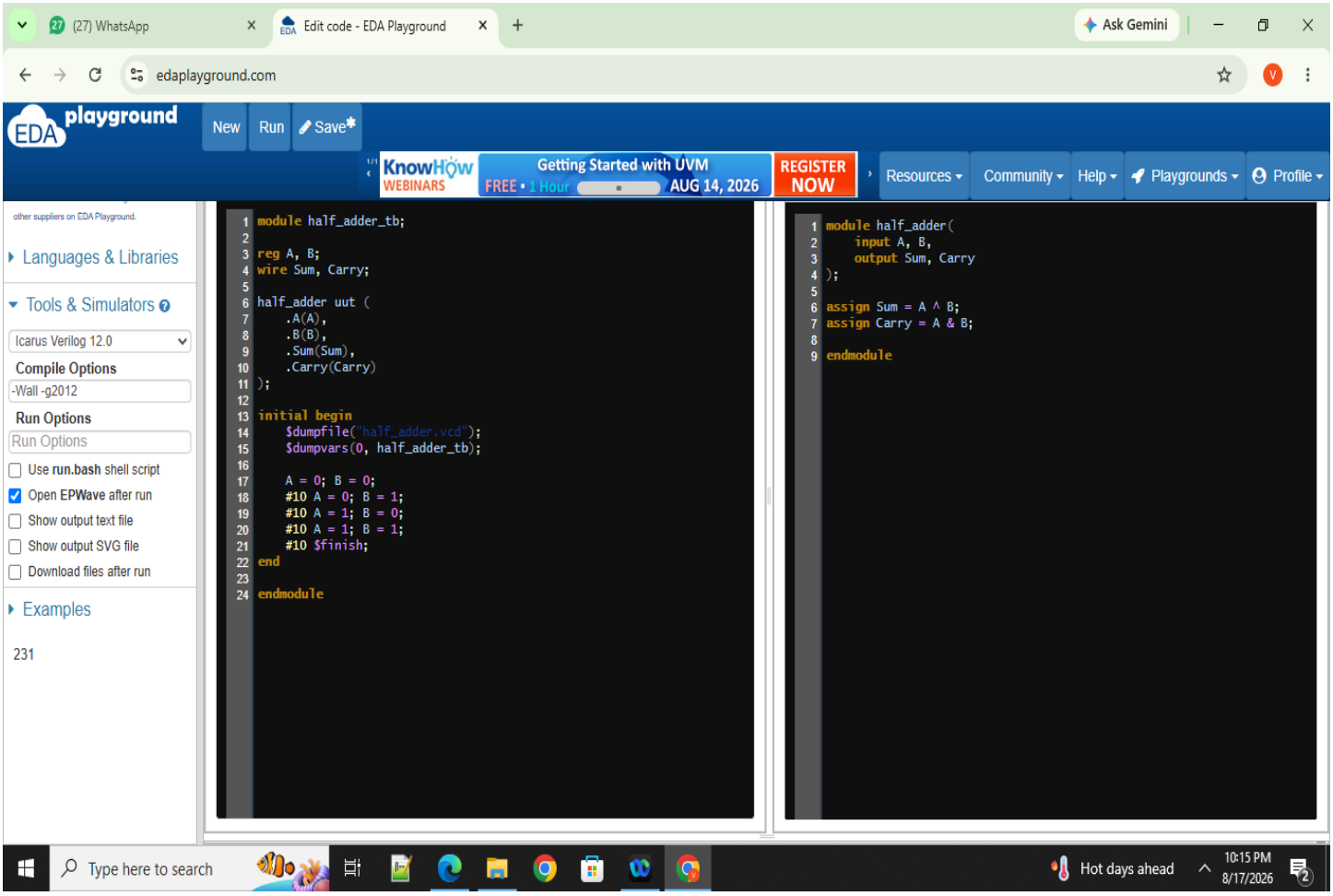


SIMULATION OUTPUT



HALF ADDER

The Half Adder was implemented using Verilog HDL to perform binary addition of two single-bit inputs. The simulation verifies the Sum and Carry outputs for all possible input combinations.

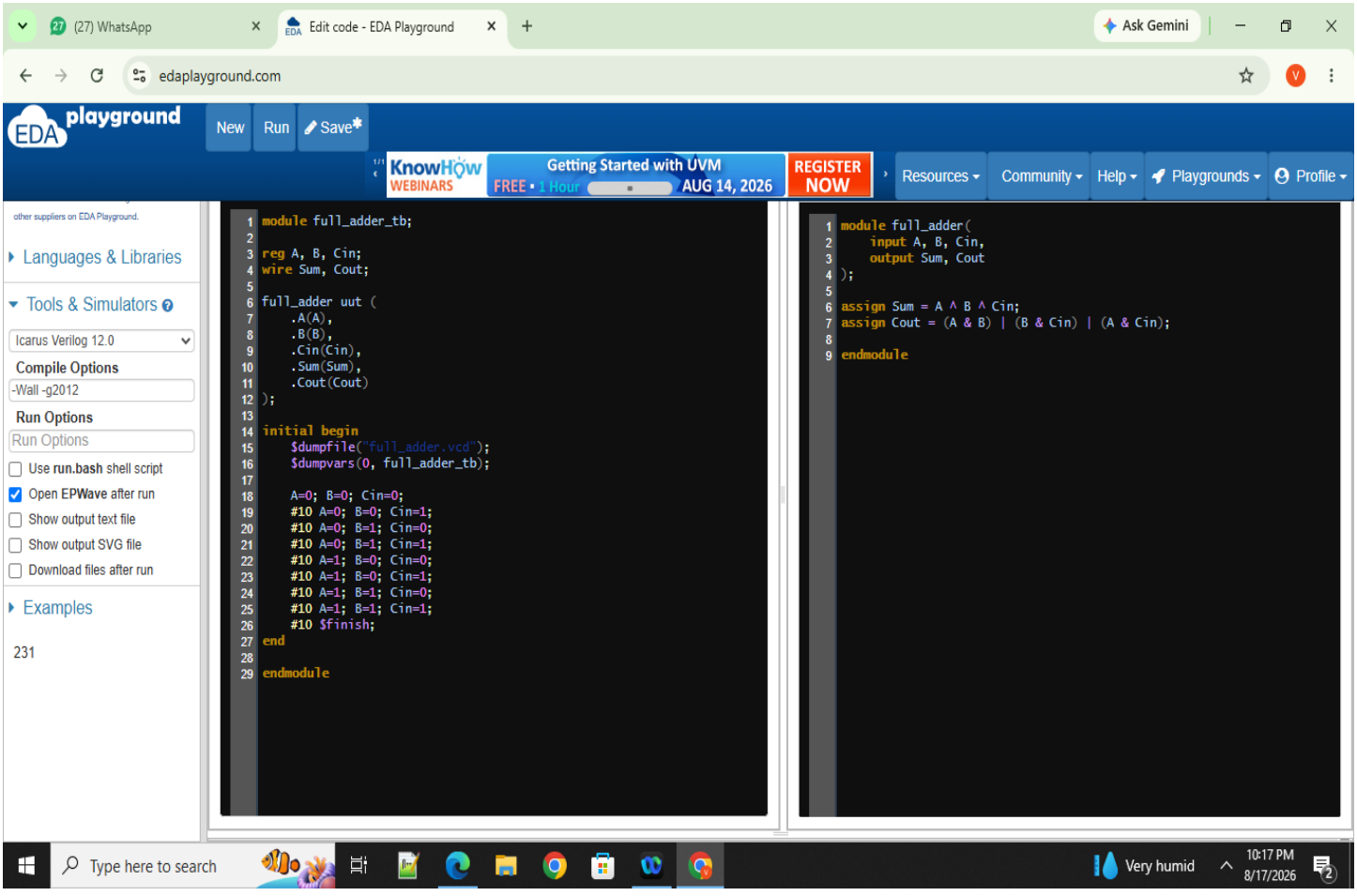


SIMULATION OUTPUT

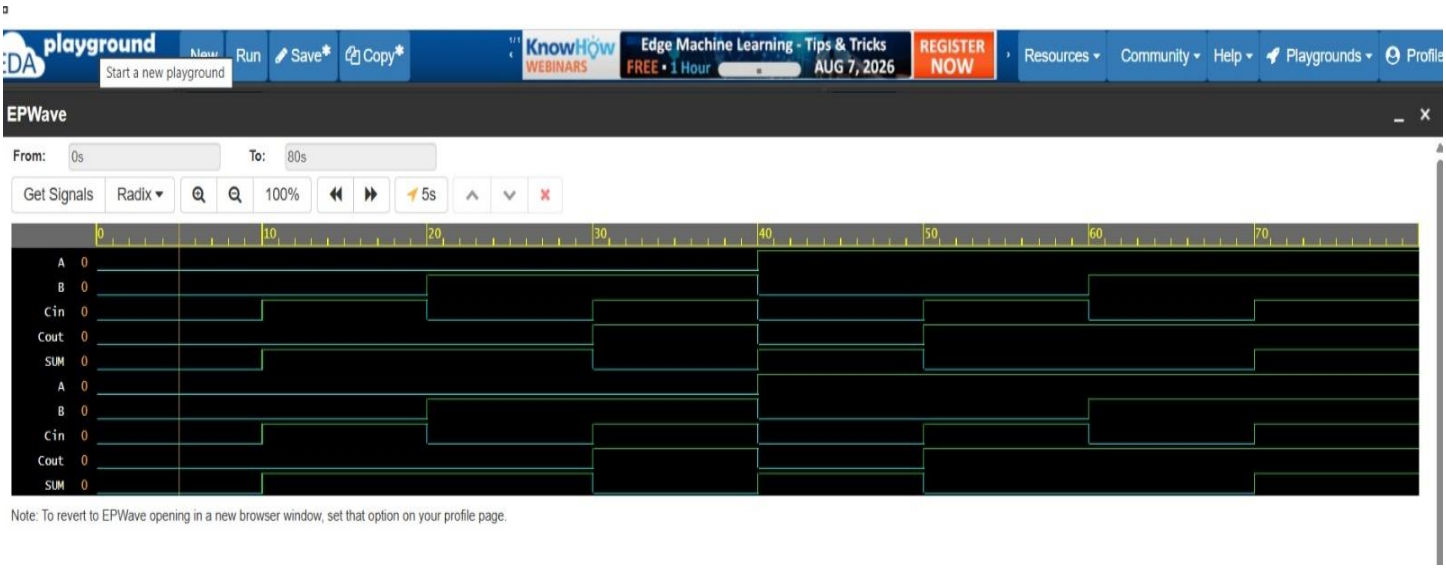


FULL ADDER

The Full Adder was implemented using Verilog HDL to add three single-bit inputs: A, B, and Cin. The simulation verifies the Sum and Carry outputs for all possible input combinations.



SIMULATION OUTPUT



CONCLUSION

The Logic Gates, Half Adder, and Full Adder were successfully designed and implemented using Verilog HDL. Their functionality was verified through simulation, and the obtained outputs matched the expected results.